Summary in Graph

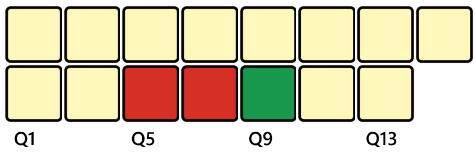
Exam Summary (GO Classes CS/DA Test Series 2026 | Databases | Topic Wise Test 2)

Qs. Attempted:	**5** (0 + 5)	**Correct Marks:**	**2** (0 + 2)
Correct Attempts:	**1** (0 + 1)	**Penalty Marks:**	**1.33** (0 + 1.33)
Incorrect Attempts:	**4** (0 + 4)	**Resultant Marks:**	**0.67** (0 + 0.67)

Total Questions:	**15** (5 + 10)
Total Marks:	**25** (5 + 20)
Exam Duration:	**45 Minutes**
Time Taken:	**45 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (157) out of 1567 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

NULL values in relational model can be used for all except which one of the following scenarios?

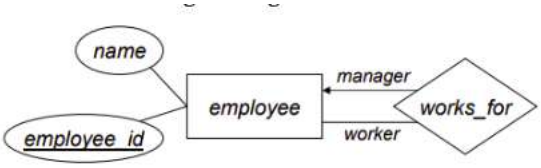
A. To allow duplicate tuples in the table by filling the primary key column(s) with NULL.
B. To avoid confusion with actual legitimate data values like 0 for integer columns and ‘ ’ (the empty string) for string columns.
C. To leave columns in a tuple marked as “unknown” when the actual value is unknown.

D. To fill a column in tuple when that column does not really "exist" for that particular tuple

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider the following ER diagram :



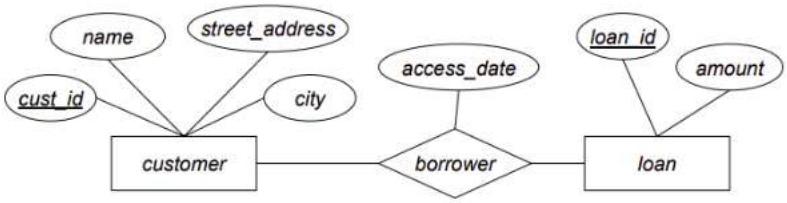
Manager-to-worker mapping is one-to-many.

We convert this ER diagram into the relational model. Minimum how many relations are needed in the relational model for this ER diagram, if we do not want null values in our database?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 22min 19sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the following ER diagram :



"borrower" mapping cardinality is many-to-many. While converting the ER diagram into a relational DBMS model, the relationship set "borrower" is converted into a relation

```
borrower(cust_id, loan_id, access_date)
```

, what is the primary key of "borrower"?

A. `cust_id`

B. `loan_id`

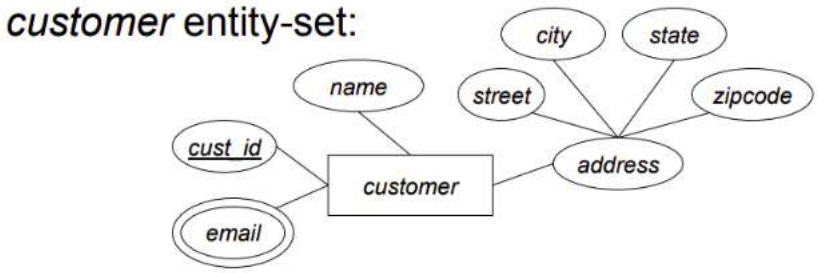
C. `(cust_id, loan_id)`

D. `(cust_id, loan_id, access_date)`

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider the following ER diagram showing a Customer entity:

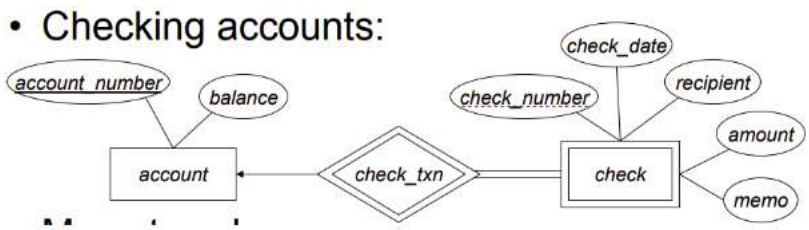


"Email" is a multi-valued attribute, So, A customer may have any number of email ids (possibly zero as well). We convert this ER diagram into the relational model. Minimum how many relations are needed in the relational model for this ER diagram?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider the following ER diagram for Checking Accounts:



We convert this ER diagram into the relational model. Minimum how many relations are needed in the relational model for this ER diagram?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

In a relation, R attribute B of R is "fully functional dependent" on an attribute or set of attribute A of R if B is functional dependent on A but not functional dependent on any proper subset of A.
Consider a relation S in which every non-prime attribute is "fully functional dependent" on all candidate keys.
What is necessarily true for S?

A. S is in $2\mathrm{NF}$.
B. S is in $3\mathrm{NF}$.
C. S is in BCNF.
D. S is not in BCNF.

Your Answer: B | Correct Answer: A | Incorrect | Time taken: 02min 13sec | Discuss

Q #7 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

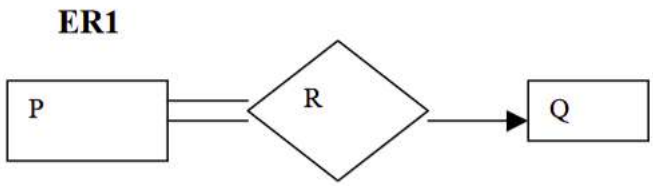
Which of the following is/are correct rules in the context of functional dependencies over a database?

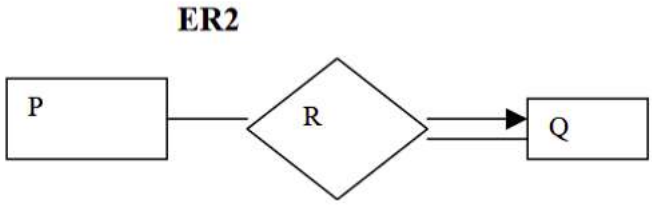
A. If α is a set of attributes and $\beta \subseteq \alpha$, then $\alpha \rightarrow \beta$ always holds.
B. If $\alpha \rightarrow \beta$ holds, and γ is a set of attributes, then $\gamma\alpha \rightarrow \gamma\beta$ holds.
C. If $\alpha \rightarrow \beta$ holds, and $\gamma\beta \rightarrow \delta$ holds, then $\alpha\gamma \rightarrow \delta$ holds.
D. If $\alpha \rightarrow \beta$ holds, and $\beta \rightarrow \gamma$ holds, then $\alpha \rightarrow \gamma$ holds.

Your Answer: A;C;D | Correct Answer: A;B;C;D | Incorrect | Time taken: 02min 57sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following two ER diagrams -- ER1 and ER2.





The statement that the number of entities in an entity set P must be greater than or equal to the number of entities in an entity set Q holds for

A. ER1 but not ER2
B. ER2 but not ER1
C. Both ER1 and ER2
D. Neither ER1 nor ER2

Your Answer: A | Correct Answer: B | Incorrect | Time taken: 03min 52sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following schema.

```
create table T (
    c integer primary key,
    d integer);
```

```
create table S (
    b integer primary key,
    c integer references t(c) on delete cascade);
```

```
create table R (
    a integer primary key,
    b integer references s(b) on delete set null);
```

Suppose the current contents of R, S, and T are as follows.

R	
A	B
1	1
2	2

S	
B	C
1	1
2	1

T	
C	D
1	1
2	1

After executing the command:

```
delete from T;
```

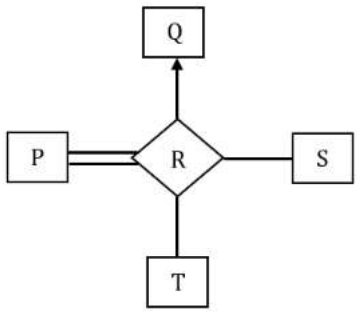
What tuples will R contain? (By the above command, all records in the table T will be deleted!)

A. R will not be changed.
B. $(1, \mathrm{NULL})$ and $(2, 2)$.
C. $(1, \mathrm{NULL})$ and $(2, \mathrm{NULL})$.
D. $(2, 2)$ only.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following ER diagram :



Primary keys of the entity set $\mathrm{P}, \mathrm{Q}, \mathrm{S}, \mathrm{T}$ are $\mathrm{M}, \mathrm{N}, \mathrm{O}, \mathrm{K}$ respectively. While converting the ER diagram into relational DBMS model, the relationship set R is converted into a relation $\mathrm{R}(\mathrm{M}, \mathrm{N}, \mathrm{O}, \mathrm{K})$, what is the primary key of R?

A. M
B. KNO
C. MKO
D. N

Your Answer: C | Correct Answer: C | Correct | Time taken: 08min 48sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following five ER diagrams - $\mathrm{ER1}$ to $\mathrm{ER5}$.

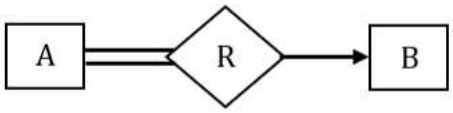
ER1:



ER2:



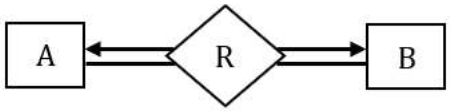
ER3:



ER4:



ER5:



For which of these ER diagrams the following is true :
"The number of entities in an entity set B must be greater than or equal to the number of entities in an entity set A"

A. $\mathrm{ER1}$, $\mathrm{ER2}$ only
B. $\mathrm{ER5}$ only
C. $\mathrm{ER3}$, $\mathrm{ER5}$ only
D. $\mathrm{ER4}$, $\mathrm{ER5}$ Only

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

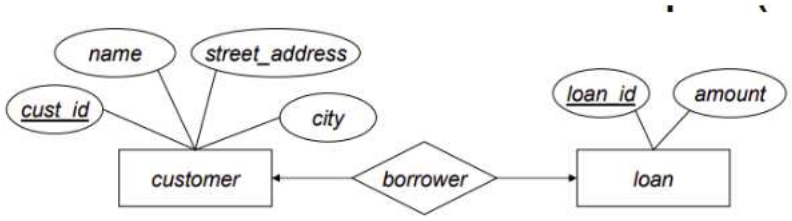
Under a relational database system, in the most general case, if table R has a foreign key (FK) constraint referencing table S, then, via the FK,

A. each tuple in R is related to zero or more tuples in S.
B. each tuple in R is related to zero or one tuple in S.
C. each tuple in R is related to exactly one tuple in S.
D. each tuple in R is related to one or more tuples in S.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the following ER diagram:



There is One-to-one mapping between customers, loans.

Assume that the Participation of loan in borrower is total.

We convert this ER diagram into the relational model. What will be the correct relational model design for this ER diagram, if we do not want null values in our database?

A.

```
customer(*cust_id, name, street_address, city) ;
loan (*loan_id, amount ) ;
borrower(*cust_id, loan_id ) ;
```

B.

```
customer(*cust_id, name, street_address, city) ;
loan (*loan_id, cust_id, amount ) ;
```

C.

```
customer(*cust_id, name, street_address, city) ;
loan (*loan_id, amount ) ;
borrower(cust_id, *loan_id ) ;
```

D.

```
customer(*cust_id, loan_id, name, street_address, city) ;
loan (*loan_id, amount ) ;
```

Your Answer: B;D | Correct Answer: A;B;C | Incorrect | Time taken: 05min 33sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider a relation R with $2d$ attributes (d is an integer ≥ 2) that are named $\mathrm{A}_1, \mathrm{A}_2, \mathrm{A}_3, \ldots, \mathrm{A}_{2d}$. There is a set F of $2d$ FDs on $\mathrm{R} : \mathrm{A}_i \rightarrow \mathrm{A}_{1+(i+1) \bmod 2d}$, for $i = 1 \ldots 2d$. (Here, mod is the modulo operator).

For example, suppose $d = 2$, the attributes are $\mathrm{A}_1, \mathrm{A}_2, \mathrm{A}_3, \mathrm{A}_4$ and
$\mathrm{F} = \{\mathrm{A}_1 \rightarrow \mathrm{A}_3, \mathrm{A}_3 \rightarrow \mathrm{A}_1, \mathrm{A}_2 \rightarrow \mathrm{A}_4, \mathrm{A}_4 \rightarrow \mathrm{A}_2\}$

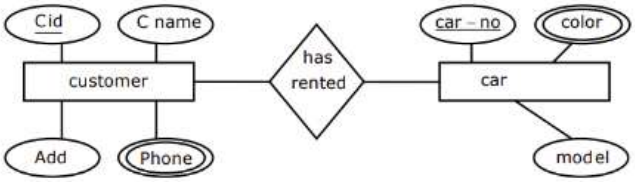
How many keys are present in R?

A. d
B. $d + 2$
C. $2d$
D. d^2

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Databases

The ER diagram shown below depicts a car rental scheme, where every customer can take more than 1 car for rent, or 1 car may be rented to multiple customers.



If the above ER diagram is mapped to a relational model, to correctly depict the above scenario, the total number of required relations is _____

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

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